

DRL-Based Robust Adaptive Traffic Signal Control with Low V2I Penetration Rate under Mixed Environment

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Abstract—Deep Reinforcement Learning (DRL) has become a widely adopted approach for adaptive traffic signal control, owing to its flexibility and effectiveness. However, DRL faces dimensionality and convergence challenges in large-scale multi-agent settings. Additionally, simplified representations of traffic conditions limit adaptability under mixed environments with pedestrians and weather factors. Moreover, in Vehicular Ad-Hoc Networks (VANETs) scenarios, low Vehicle-to-Infrastructure (V2I) penetration exacerbates the complexity of traffic observation and coordination. In this paper, we propose a DRL-based robust adaptive traffic signal control with a low V2I penetration rate under mixed environment (DRATSC). Specifically, DRATSC builds upon the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework, modeling intersections as autonomous agents and leveraging V2I to obtain real-time data. An attention mechanism is employed to dynamically focus on neighboring intersections, optimizing traffic signal coordination. The method overcomes low V2I penetration through scenario reconstruction and models mixed traffic environments by incorporating weather impacts, pedestrian interactions, and diverse action spaces. Extensive experiments conducted in the Simulation of Urban Mobility (SUMO) platform show that the proposed method improves overall performance by 40% over conventional approaches and outperforms IPDALight across most scenarios.

Index Terms—Traffic Signal Control, Deep Reinforcement Learning, VANETs

I. INTRODUCTION

In recent years, Deep Reinforcement Learning (DRL) [1] has emerged as a promising method for Intelligent Traffic Signal Control (ITSC). DRL agents learn optimal traffic control policies through environment interaction, enabling real-time adaptive decisions that enhance urban traffic efficiency.

However, DRL-based traffic signal control still faces three challenges: 1) The exponential expansion of action spaces with increasing intersection numbers leads to severe scalability and coordination difficulties in large-scale multi-agent scenarios [2]. 2) Low Vehicle-to-Infrastructure (V2I) penetration rates lead to incomplete or inaccurate traffic data [3]. 3) Adverse weather and unpredictable pedestrian behavior disrupt signal control strategies [4].

Firstly, to address the challenges of multi-signal control in large-scale traffic networks, researchers have enhanced traditional DRL approaches by incorporating advanced neural architectures (e.g., Long Short-Term Memory (LSTM),

attention mechanisms) to achieve coordinated signal control across intersections. Wu et al. [5] proposed a novel multi-agent recurrent deep deterministic policy gradient (MARDDPG) algorithm to coordinate the traffic light controllers. However, their study noted that the method failed to converge in scenarios involving 16 intersections. Colight [6] is the first to use graph attentional networks in the setting of reinforcement learning for traffic signal control and to conduct experiments on the large-scale road network with hundreds of traffic signals. Some research also focuses on decomposing the original challenging Multi-Agent Reinforcement Learning (MARL) task into multiple subproblems to optimize model complexity [2]. However, whether employing only adjacent intersection communication or area-based control, both approaches require receiving information from other traffic signals during the control phase, making them susceptible to transmission delays and packet loss [7].

Secondly, for the issue of incomplete traffic data acquisition, this paper primarily focuses on obtaining real-time traffic information through Vehicular Ad-Hoc Networks (VANETs). VANETs support vehicle-to-vehicle (V2V) and V2I [8] communication for real-time traffic data collection, including vehicle count, speed, direction, route, and driver preferences [9]. Some existing studies [10] have already utilized information provided by Vehicle-to-Everything (V2X) as state inputs for traffic signal control strategies. However, many approaches often overlook the fact that not all vehicles are equipped with VANETs technology, and that even for connected vehicles, communication reliability is far from guaranteed. For example, when considering a realistic signal propagation model and a moderate distance of 300 meters between transmitter and receiver, the application-level reliability of the Dedicated Short-Range Communications (DSRC)-based communication is no more than 55% [11]. Therefore, complete and reliable traffic information is essential for ensuring both robustness and efficacy in traffic signal control. Some studies have employed reinforcement learning [3] and incorporated LSTM networks [5] to handle incomplete traffic data. However, these approaches lack interpretability, making it difficult to verify whether they truly reconstruct complete and accurate traffic information.

Finally, to address complex traffic environments, prior studies have identified several key factors influencing traffic dynamics. Among them, waiting time at red lights has been shown to be a major contributor to pedestrian jaywalking behavior [12]. Besides, in adverse weather conditions such

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as heavy rain or fog, traffic congestion is often exacerbated [13]. Research by Ahmed et al. [14] demonstrated that the probability of reducing speed more than 5 kph below the speed limits were 23% and 29% higher in light rain and heavy rain, respectively, which will cause differences in vehicle behavior compared to daily training data. Thus, traffic optimization should simultaneously address pedestrian flows and external factors like weather variations. While some studies have incorporated pedestrian considerations into traffic signal control [4], few have accounted for weather factors in their traffic signal control frameworks.

To the best of our knowledge, the existing methods cannot simultaneously address the three challenges mentioned above. Therefore, we introduce a novel DRL-based approach for robust adaptive traffic signal control under mixed environments, considering pedestrians, weather, and the increased complexity due to low V2I penetration rates.

The main contributions of this paper are as follows:

1) We propose a new method that enhances the MADDPG architecture by integrating a multi-head attention mechanism into the critic network during training. The attention module computes dynamic weights for directly connected neighboring intersections to capture their influence on signal control. In the inference phase, the actor network operates solely on local state information, eliminating the need for inter-agent communication and thus avoiding communication delays.

2) We propose a deep learning-based Scenario Reconstruction Module to address the challenge of incomplete traffic data caused by low V2I penetration rates. This module effectively reconstructs missing vehicle information to provide complete environmental observations for our framework, thereby improving the quality of input data and enhancing the performance of intersection control strategies.

3) We adopt a comprehensive simulation environment integrating pedestrian traffic and adverse weather conditions to enhance safety and control efficiency. Pedestrian flows are incorporated into signal coordination to reduce waiting time, while weather-induced variations in vehicle speed and headway are embedded in the state space to improve robustness and generalization.

To validate the proposed method, we conducted extensive experiments in seven traffic scenarios using the Simulation of Urban Mobility platform (SUMO). Results show that the method consistently outperforms baselines in vehicle and pedestrian waiting time as well as vehicle travel time.

The remainder of this paper is organized as follows. Section II reviews related work. Section III provides the preliminaries of the research content. Section IV introduces the method framework and details. Section V shows the experimental details and results. Section VI analyzes the model performance. Finally, this paper is concluded in Section VII.

II. RELATED WORK

A. Conventional Traffic Signal Control

Early research in traffic signal control primarily focused on rule-based systems that relied on statistical traffic data [15], such as SCOOT [16], SCATS [17] and Max Pressure

(MP) [18]. While these approaches provided certain benefits, a significant limitation was their inability to dynamically adapt to real-time traffic conditions.

Therefore, some researchers constructed mathematical models with logic constraints to cope with more complex traffic environment. Zhang et al. [19] proposed a traffic signal scheduling strategy considering both pedestrian and vehicle delays in urban systems. They model pedestrian and vehicle flows, formulating the problem as mixed-integer quadratic programming (MIQP). Villagra et al. [20] proposed two versions of a Cellular Genetic Algorithm (cGA) to optimize traffic light cycles, enhancing vehicle flow without additional infrastructure or driver applications. Zhang et al. [21] proposed a novel model for the dynamic behavior of heterogeneous traffic networks, where signalized intersections were modeled using piecewise linear approximations, and non-signalized intersections were described using a platooning-like First-in-First-out (FIFO) model. The resulting traffic light control problem was formulated as a mixed-integer programming problem.

Although simulations indicate that these methods could outperform the fixed-time and conventional actuated control strategies, their effectiveness in dynamic environments remains limited. This is primarily due to restrictive assumptions, such as uniform traffic flow over specific time intervals and idealized vehicle speed changes—conditions which are rarely met in practice.

B. DRL-based Traffic Signal Control

Recently, reinforcement learning techniques have been proposed to coordinate multi-intersection traffic signals. According to the different challenges faced, DRL-based ITSC studies can be divided into three categories: considering multi-intersection coordination, considering incomplete traffic state observation and considering comprehensive implementation factors.

Considering multi-intersection coordination. Many approaches have explored different coordination scopes, focusing either on neighbor-node influences or region-level coordination. Wei et al. [6] proposed a model, CoLight, which learned to communicate on the representations of neighboring intersections by leveraging the attention mechanism. Chen et al. [22] proposed MPLight, which employed the pressure concept to achieve region-level signal coordination. Building on these studies, Zhang et al. [23] proposed Advanced-MP by integrating traffic movement representations with CoLight and MPLight frameworks, demonstrating superior performance. KeyLight [24] is a DRL-based traffic signal control method that highlights critical information through state representation called NOV-LADLE and residual connections in Graph Neural Networks (GNNs). Liu et al. [25] proposed MAADAC, which is a multi-agent Semi-Markov decision framework integrating Graph Attention Networks (GATs) and the options paradigm for scalable traffic signal coordination. Zhao et al. [26] took the coordination of neighboring intersections into account, enabling the fine-tuning of phase duration of traffic signals to adapt to dynamic traffic situations. In addition, there are

some studies that solve the problem of multi-intersection ITSC by decomposing the idea of breaking down large problems. Tan et al. [2] implemented a two-tiered framework with regional agents and a centralized global agent to implement large-scale traffic grid signal control. Similarly, Zhou et al. [27] addressed large-scale control by reformulating it as a decentralized multi-agent problem, enabling effective multi-intersection coordination.

However, most studies assume that signal controllers can collect all instantaneous vehicular information without delay or loss. In practice, communication delays and packet loss [7] often result in missing information, particularly for distant intersections, which directly undermines traffic state estimation and signal control performance [28].

Considering incomplete traffic state observation. Most ITSC algorithms assume full vehicle detectability via cameras or loop detectors. However, these sensors are prone to hardware failures and environmental interference. With the rise of VANETs, some studies have proposed cost-effective ITSC management methods by detecting vehicles using V2I technology, such as DRLE [27], CoTV [29] and IASC [30]. Nevertheless, their effectiveness depends heavily on VANETs penetration rates.

To deal with such issues, some studies have focused on partially detected ITSC systems. Wu et al. [5] utilized LSTM to capture state information to deal with low penetration situation. Zhang et al. [3] used Deep Q-Network (DQN) to train the traffic lights agent, which proved that the system under DRL control is much better than the traditional fixed-time traffic signal, even with sparse observations. These studies usually deal with the problem of low penetration by directly using or modifying the reinforcement learning architecture, but the results are unexplainable.

Considering comprehensive implementation factors. For mixed traffic environments, numerous studies have already taken various real-world factors into account. Yazdani et al. [4] proposed “total user delays” instead of “total vehicle delays” to allocate an equitable green time to vehicles and pedestrians. Wu et al. [5] assigned different weights to different types of vehicles and considered the pedestrian waiting delay at the intersection. Liu et al. [25] effectively reduced the waiting time of high weight vehicles by weighting them. Li et al. [31] adaptively handled complex and diverse real-world intersection types in traffic light control by introducing a hierarchical mixed reinforcement learning framework with enhanced agent cooperation. However, few studies have considered weather factors.

III. PRELIMINARIES

A. VANETs and Partial Detection

VANETs facilitate wireless communication, supporting both V2V and V2I interactions [32].

V2V Communication: V2V facilitates direct vehicle-to-vehicle data exchange without relying on fixed infrastructure, enhancing real-time awareness of nearby traffic conditions.

V2I Communication: V2I supports bidirectional interaction between vehicles and roadside units (RSUs), enabling

real-time data transmission such as vehicle speed, position, and trajectory. In ITSC systems, V2I plays a critical role in collecting vehicle-level data for adaptive signal optimization.

However, VANETs functionality depends on vehicles being equipped with onboard communication units (OBUs). In practice, achieving full penetration (100% vehicle connectivity) in VANETs is unrealistic. Consequently, vehicle information is partially available, resulting in incomplete or inconsistent observations.

Fig. 1 shows the Partially Detected ITSC System [3]. Detected vehicles (blue) are equipped with OBUs and communicate with RSUs, providing real-time data. In contrast, undetected vehicles (white) lack OBUs and are invisible to the system, leading to incomplete traffic state observations.

To address this challenge, we propose a deep learning-based module that captures the spatiotemporal dependencies among adjacent vehicles, enabling the reconstruction of undetected vehicle information and improving the robustness of ITSC under low-penetration conditions.

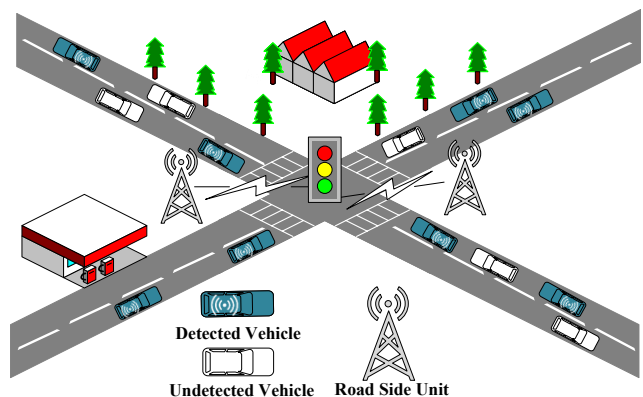


Fig. 1: Partially Detected ITSC System

B. Multi-Agent Deep Deterministic Policy Gradient Algorithm

Multi-Agent Deep Deterministic Policy Gradient (MADDPG) is an algorithm for multi-agent reinforcement learning (MARL) problems [33]. Its core strength is the Centralized Training with Decentralized Execution (CTDE) paradigm: during training, each agent's critic has access to the global state and joint actions, while during execution, each actor makes decisions based only on local observations.

This framework is well suited to multi-intersection traffic signal control. Centralized critics capture spatial interdependencies among intersections, enabling agents to learn coordinated strategies that reduce network-level congestion. During deployment, each signal controller relies only on local traffic observations, ensuring scalability and practical feasibility in real-world systems where global information is unavailable. As shown in Fig. 2, each agent uses an independent actor for real-time decisions, while training is guided by critics leveraging system-wide information.

The MADDPG algorithm extends the single-agent Deep Deterministic Policy Gradient (DDPG) framework, adapting its distinctive actor-critic architecture to multi-agent systems

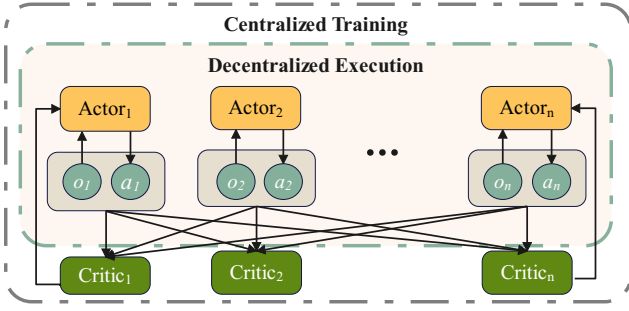


Fig. 2: Centralized Training with Decentralized Execution Structure of MADDPG

through a centralized training with decentralized execution paradigm [33]. Further, DDPG is a deep learning-based extension of Deterministic Policy Gradient (DPG), using neural networks for function approximation in continuous action spaces. DPG uses deterministic policies, meaning that for a given state, the policy outputs a specific action without any randomness. Therefore, the deterministic policy can be expressed as $a = \mu(o)$, where $\mu(o)$ denotes the policy function mapping the agent's observation o to a specific action a .

For a system with N agents employing continuous policies μ_{θ_i} (abbreviated as μ_i) parameterized by θ_i , the policy gradient is computed as:

$$\nabla_{\theta_i} J(\mu_i) = \mathbb{E}_{x, a \sim \mathcal{D}} \left[\nabla_{\theta_i} \mu_i(o_i) \nabla_{a_i} Q_i^{\mu}(x, a_1, \dots, a_N) \Big|_{a_i = \mu_i(o_i)} \right] \quad (1)$$

where $\nabla_{\theta_i} J(\mu_i)$ represents the gradient of the objective function $J(\mu_i)$ with respect to the parameters θ_i of the policy μ_i . $Q_i^{\mu}(x, a_1, \dots, a_N)$ is a centralized action-value function, approximated by a neural network that takes the actions of all agents $a_1, \dots, a_N$ and state information x as input, then outputs the Q-value for agent i . In a simple case, x could consist of the observations of all agents, $x = (o_1, \dots, o_N)$. The experience replay buffer $\mathcal{D}$ contains the tuples $(x, x', a_1, \dots, a_N, r_1, \dots, r_N)$, recording experiences of all agents. The centralized action-value function Q_i^{μ} is updated as:

$$\begin{cases} \mathcal{L}(\omega_i) = \mathbb{E}_{x, a, r, x'} \left[(Q_i^{\mu}(x, a_1, \dots, a_N) - y)^2 \right] \\ y = r_i + \gamma Q_i^{\mu'}(x', a'_1, \dots, a'_N) \Big|_{a'_j = \mu'_j(o'_j)} \end{cases} \quad (2)$$

where $\mu' = \{\mu_{\theta'_1}, \dots, \mu_{\theta'_N}\}$ is the set of target policies with delayed parameters θ'_i .

It is worth mentioning that the designed action space in this paper is discrete. But DDPG requires the agent's actions to be continuous. Therefore, the Gumbel-Softmax method [34] is used to approximate sampling from the discrete distribution.

In Gumbel-Softmax method, a reparameterization factor g_i is introduced, which is a sample from a Gumbel(0,1) distribution:

$$g_i = -\log(-\log u), \quad u \sim \text{Uniform}(0, 1) \quad (3)$$

The Gumbel-Softmax sampling can be formulated as:

$$y_i = \frac{\exp((\log a_i + g_i)/\tau)}{\sum_{j=1}^k \exp((\log a_j + g_j)/\tau)}, \quad i = 1, \dots, k \quad (4)$$

where a_i is an action probability value obtained through a neural network. The discrete action is selected as $z = \arg \max y_i$, while the continuous output y maintains differentiability for gradient backpropagation. The temperature parameter τ governs the approximation quality of the Gumbel-Softmax distribution to its discrete counterpart: as $\tau \rightarrow 0^+$, the distribution converges to the one-hot encoding of $\arg \max_i (\log a_i + g_i)$, whereas increasing τ causes the distribution to approach a uniform probability mass function.

IV. THE PROPOSED METHOD

A. Notations

Table I presents the key mathematical notations adopted in this paper. The state representation $S_{t,i}$ is formed from the reconstructed observation $\hat{O}_{t,i}$, incorporating traffic matrices of position (P_i), velocity (V_i), and other traffic-related features. Here, i indexes agents and t denotes time steps.

Note that in scenarios without partial detection concerns, the raw observation $O_{t,i}$ and the reconstructed observation $\hat{O}_{t,i}$ are identical, as no reconstruction is needed.

TABLE I: Notations

Notation	Description
P_i	Binary position matrix of agent i .
V_i	Velocity matrix of agent i .
Q_{ds}	Matrix storing the number of queued vehicles on downstream roads.
H	Matrix of average headway distances.
P_{ped}	Matrix storing the number of pedestrians waiting to cross in each direction.
$O_{t,i}$	Raw observation of agent i at time t , possibly incomplete due to low V2I penetration.
$\hat{O}_{t,i}$	Reconstructed observation of agent i at time t , output from the Scenario Reconstruction Module.
$S_{t,i}$	State of agent i at time step t .
a_i	Action taken by agent i .
r_i	Reward received by agent i .
w_{veh}	Total waiting time of vehicles in the current phase.
p_{veh}	Number of vehicles that passed through the intersection in the current phase.
w_{ped}	Total waiting time of pedestrians in the current phase.
α, β, σ	Weight coefficients in the reward function.
Q_i^{μ}	Centralized action-value function for agent i under policy μ .
θ_i	Parameters of the policy network (Actor) for agent i .
ω_i	Parameters of the value network (Critic) for agent i .
τ	Soft update coefficient for target networks.
leg_j	The approach information of intersection i directly connected to intersection j .
m_j	Message tuple received from neighboring intersection j , containing vehicle count and action.

B. Agent Design

In our method, each intersection is modeled as an autonomous agent capable of observing local traffic conditions and communicating with neighbor intersections in different directions. For each agent, the state, action, and reward are set as follows.

1) *State*: For each agent, the state is represented by two complementary traffic matrices derived from VANETs data: position and velocity. As illustrated in Fig. 3, each road is uniformly discretized along the travel direction, where occupied cells in the position matrix are marked as 1 and empty cells as 0. The velocity matrix records the corresponding vehicle speeds. The position and velocity matrices of agent i are denoted by P_i and V_i , are defined as:

$$P_i = \begin{bmatrix} p_{1,1} & p_{1,2} & \cdots & p_{1,n} \\ p_{2,1} & p_{2,2} & \cdots & p_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{l,1} & p_{l,2} & \cdots & p_{l,n} \end{bmatrix} \quad (5)$$

$$V_i = \begin{bmatrix} v_{1,1} & v_{1,2} & \cdots & v_{1,n} \\ v_{2,1} & v_{2,2} & \cdots & v_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{l,1} & v_{l,2} & \cdots & v_{l,n} \end{bmatrix}$$

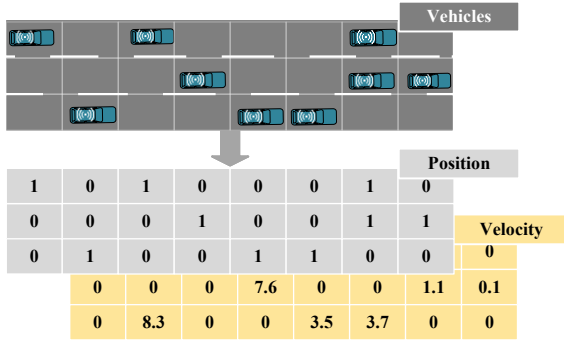


Fig. 3: State: Position and Speed Matrix

Additionally, the number of queued vehicles on each downstream road is represented by a matrix. As shown in Fig. 4, downstream information can be observed locally, as it corresponds to the opposing roads of the same intersection. Incorporating downstream information allows the agent to recognize congestion in the direction it plans to release traffic and avoid adding more vehicles to already congested downstream roads. The downstream information matrix Q_{ds} is defined as follows:

$$Q_{ds} = [q_{ns}, q_{ew}, \cdots] \quad (7)$$

where q_{ns} is the number of vehicles on the downstream roads in the north-south direction. q_{ew} is the number of vehicles on the downstream roads in the east-west direction. The content of Q_{ds} can be adjusted according to the design of the actions.

Under adverse weather conditions such as heavy rain or fog, traffic congestion tends to worsen [13]. Drivers typically reduce their speed and increase following distances in poor weather [35]. To capture these behavioral variations, the state definition incorporates the headway distance H , which providing a more robust description of traffic conditions:

$$H = [H_1, H_2, \dots, H_L] \quad (8)$$

where $H_l = \frac{1}{M_l} \sum_{i=1}^{M_l} h_{l,i}$ is the average headway distance of vehicles on lane l , M_l is the number of vehicles on that lane,

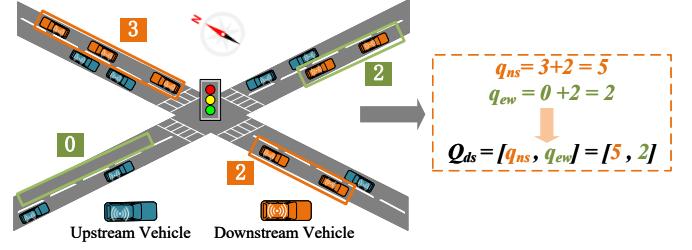


Fig. 4: The Number of Queued Vehicles on Downstream Roads

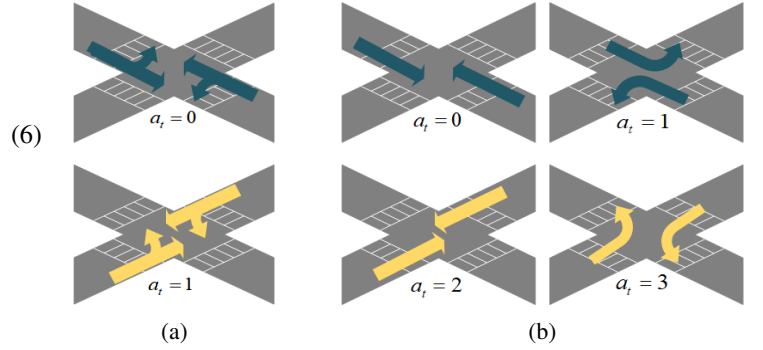


Fig. 5: Action Schemes: (a) Two-Phase Scheme, (b) Four-Phase Scheme

and $h_{l,i}$ denotes the headway distance of the i -th vehicle on lane l . L is the total number of lanes.

Moreover, pedestrians on sidewalks should not be overlooked. Failing to account for pedestrians and causing excessive waiting times can significantly increase the likelihood of jaywalking [12]. Therefore, the matrix P_{ped} is also used to store the number of pedestrians near the intersection, which is given by:

$$P_{ped} = [p_{ns}, p_{ew}] \quad (9)$$

where p_{ns} is the number of pedestrians crossing in the north-south direction. p_{ew} is the number of pedestrians crossing in the east-west direction.

In summary, the state matrix of agent i at time step t can be expressed as:

$$S_{i,t} = (P, V, Q_{ds}, H, P_{ped}) \quad (10)$$

2) *Action*: In traffic signal control systems, four-leg intersections typically employ either two-phase or four-phase timing schemes, each offering distinct advantages depending on operational requirements. As illustrated in Fig. 5a, the two-phase scheme alternates between two primary movement groups (e.g., north-south and east-west through movements), while Fig. 5b demonstrates the four-phase scheme with dedicated phases for protected left-turn movements. While the two-phase scheme demonstrates superior performance in minimizing overall vehicle delays, the four-phase alternative proves more effective in mitigating left-turn lane congestion, particularly during peak traffic conditions [36]. Therefore, our method will conduct experiments in two action spaces separately.

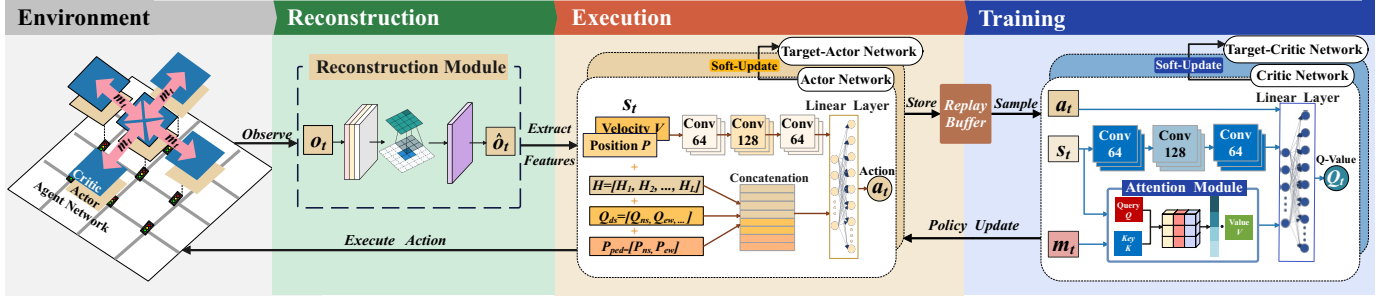


Fig. 6: DRATSC Network Framework

Algorithm 1: DRATSC

Input: E : number of episodes; T : number of timesteps; N : number of agents

- 1 **Initialize:** critic network Q_ω , actor networks μ_θ ;
- 2 target networks Q'_ω, μ'_θ .
- 3 **for** $e = 1$ **to** E **do**
- 4 Reset environment;
- 5 **for** $t = 1$ **to** T **do**
- 6 // Action selection
- 7 **for** *each agent* i **do**
- 8 Obtain original observation o_i ;
- 9 Get reconstructed observation $\hat{o}_i$;
- 10 Extract features to form state s_i ;
- 11 Receive messages m_i from surrounding agents;
- 12 $a_i \leftarrow \mu_{\theta_i}(s_i)$;
- 13 Form joint action $\mathbf{a} = (a_1, \dots, a_N)$;
- 14 // Environment interaction
- 15 Execute joint action $\mathbf{a}$;
- 16 Obtain rewards $\mathbf{r}$ and new observations $\mathbf{o}'$;
- 17 **for** *each agent* i **do**
- 18 Get reconstructed observation $\hat{o}'_i$;
- 19 Extract new state s'_i ;
- 20 Receive new messages m'_i from surrounding agents;
- 21 Form joint tuples $\mathbf{s} = (s_1, \dots, s_N)$,
- 22 $\mathbf{m} = (m_1, \dots, m_N)$, $\mathbf{s}' = (s'_1, \dots, s'_N)$,
- 23 $\mathbf{m}' = (m'_1, \dots, m'_N)$;
- 24 Store transition $(\mathbf{s}, \mathbf{m}, \mathbf{a}, \mathbf{r}, \mathbf{s}', \mathbf{m}')$ in $\mathcal{D}$;
- 25 // Network update
- 26 Sample a minibatch from $\mathcal{D}$;
- 27 **for** *each agent* i **do**
- 28 Update critic network Q_ω ;
- 29 Update actor network μ_θ ;
- 30 Soft update target networks Q'_ω, μ'_θ ;

The transition interval between actions is set to a minimum duration Δt , adjusted according to map-specific traffic characteristics. If the selected action remains unchanged after this interval, no additional yellow phase is introduced; otherwise, a yellow-light transition is applied.

3) *Reward*: The reward function for agent i is designed to jointly promote traffic efficiency and fairness. It integrates three measurable components: total vehicle waiting time w_{veh} ,

total pedestrian waiting time w_{ped} , and the number of vehicles passing through the intersection p_{veh} in the current phase. The waiting-time terms penalize excessive delays for both vehicles and pedestrians, while the throughput term rewards efficient traffic discharge. By combining these objectives, the agent learns to balance intersection efficiency with service equity under varying traffic demands.

These components are weighted by hyperparameters α , β , and σ respectively, controlling their relative contributions to the overall optimization objective. The resulting reward function is given by:

$$r_i = -(\alpha w_{veh} + \beta w_{ped} - \sigma p_{veh}) \quad (11)$$

where the leading negative sign transforms the cost minimization into a reward maximization problem.

C. Framework of the Method

The framework of our method is illustrated in Fig. 6. DRATSC is built upon the MADDPG algorithm, where each intersection is modeled as an individual agent equipped with an Actor–Critic architecture and corresponding target networks to stabilize parameter updates. After acquiring raw traffic information from the environment, incomplete observations are first processed by the Scenario Reconstruction Module to generate reconstructed environmental states. Based on these states, the actor network infers traffic signal actions, while the critic network, enhanced with an attention mechanism, evaluates joint actions by accounting for interactions among neighboring intersections. The overall learning and execution procedure of DRATSC is summarized in Algorithm 1.

Actor Network. Each intersection agent first collects real-time traffic observations via VANETs. To handle partial detection, the Scenarios Reconstruction Module (Section IV-D) recovers incomplete observations o_t to obtain reconstructed representations $\hat{o}_t$, which are then used to form the enhanced state s_t . The actor network takes s_t as input and outputs an action probability distribution, as detailed in Table II. Since it relies only on local state information, the inference process operates independently without inter-agent communication, thereby improving computational efficiency.

Critic Network. In the MADDPG framework, centralized critic training involves global states and actions, which can cause dimensional expansion and increase computational complexity. Considering that the influence of distant intersections on local decisions is limited at time t , the critic network in

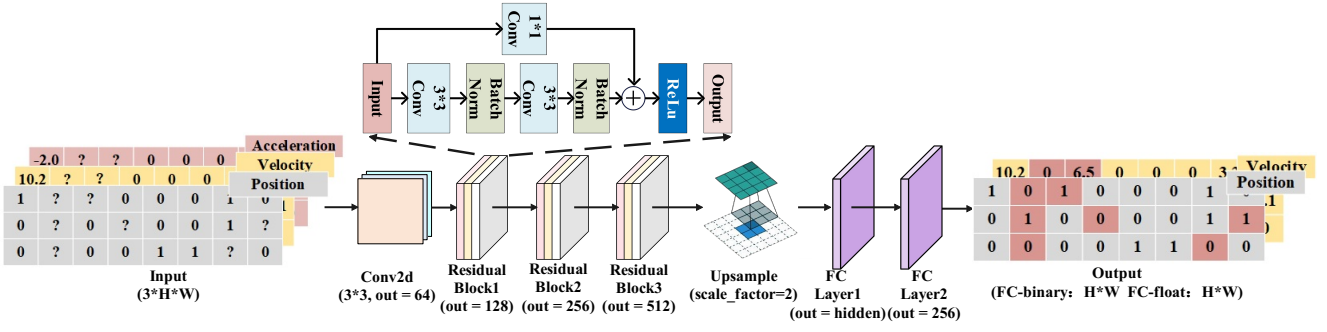


Fig. 7: Scenario Reconstruction Module

TABLE II: Actor Network Architecture

Layer	Type	Parameters
1	Conv2D	filters=64, kernel_size=4 × 4 stride=1, padding=1
2	BatchNorm2d	num_features=64
3	Conv2D	filters=128, kernel_size=4 × 4 stride=1, padding=1
4	BatchNorm2d	num_features=128
5	Conv2D	filters=64, kernel_size=4 × 4 stride=1, padding=1
6	BatchNorm2d	num_features=64
7	Flatten	–
8	Linear	in=flatten_dim, out=hidden_dim activation=LeakyReLU
9	Dropout	rate=p
10	Linear	in=hidden_dim, out=hidden_dim activation=LeakyReLU
11	Dropout	rate=p
12	Linear	in=hidden_dim, out=action_num
13	LogSoftmax	–

this paper focuses on information from directly neighboring intersections. This information can be efficiently transmitted via RSUs, as illustrated in Fig. 6, is represented as a message tuple $m = (m_1, m_2, \dots, m_k)$, where k denotes the number of neighboring agents.

The neighboring messages are integrated with the local state s_t and processed through an Attention Module (Section IV-E) to produce a weighted output. This output, together with the feature-extracted state s_t from CNN layers and the local action a_t , is fed into fully connected layers to compute the value $Q(s, a)$. The critic network shares a similar architecture with the actor network, except that its final layer outputs a scalar value rather than policy logits.

D. Scenario Reconstruction Module

Network Update. According to the above, in the critic network, the centralized action-value function is updated in Eq. 12:

$$\begin{cases} y = r + \gamma Q_i^{\mu'}(s'_i, a'_i, m'_1 \dots m'_k) \Big|_{a'_i = \mu'_i(s'_i)} \\ L(\omega_i) = \mathbb{E}_{(s, m, a, r, s', m') \sim D} \left[\left(Q_i^{\mu}(s_i, a_i, m_1 \dots m_k) - y \right)^2 \right] \end{cases} \quad (12)$$

where $L(\omega_i)$ is the loss function for agent i , (s, m, a, r, s', m') are tuples of experiences sampled from D . $s = (s_1, s_2, \dots, s_n)$

are the states of n agents. $m = ((m_1^1 \dots m_k^1), \dots, (m_1^n \dots m_k^n))$ are the messages obtained from k neighbor agents, where each m_i^j encodes the road information and selected action of neighbor i . $a = (a_1, a_2, \dots, a_n)$ are the actions of n agents. $r = (r_1, r_2, \dots, r_n)$ are the rewards of n agents. s' and m' are the state tuples and message tuples of the next stage after executing action a . Q_i^{μ} is the Q-function for agent i under policy μ . $\mu' = (\mu'_{\theta_1}, \dots, \mu'_{\theta_n})$ is the set of target policy. a'_i is the action generated by the target policy μ'_{θ_i} upon receiving the state s'_i , namely, $a'_i = \mu'_{\theta_i}(s'_i)$.

The individual policies are updated by ascent with the following gradient:

$$\begin{aligned} \nabla_{\theta_i} J(\mu_i) &= \mathbb{E}_{(s_i, a_i, m_1, \dots, m_k) \sim D} \left[\nabla_{\theta_i} \mu_i(s_i) \cdot \right. \\ &\quad \left. \nabla_{a_i} Q_i^{\mu}(s_i, a_i, m_1, \dots, m_k) \Big|_{a_i = \mu_i(s_i)} \right] \end{aligned} \quad (13)$$

where $\nabla_{\theta_i} J(\mu_i)$ is the gradient of the deterministic policy μ_i , with the goal of maximizing the expected return $J(\mu_i)$. The policy $\mu_i(s_i)$ outputs the deterministic action of agent i given its local observation s_i .

The target network is updated using soft updates, and the update formulas for the target-actor network and the target-critic network are in Eq. 14:

$$\begin{cases} \omega'_i \leftarrow \tau \omega_i + (1 - \tau) \omega'_i \\ \theta'_i \leftarrow \tau \theta_i + (1 - \tau) \theta'_i \end{cases} \quad (14)$$

where ω is the parameter of critic network, and θ is the parameter of actor network. ω' and θ' are the parameters of the corresponding target network, τ is a hyperparameter.

The dynamic traffic light control system requires real-time traffic information. However, due to limitations in equipment and weather conditions, vehicle information may only be partially detected, as illustrated in Fig. 1. To address this issue, this paper proposes a Scenario Reconstruction Module that recovers incomplete traffic states.

As shown in Fig. 7, the input consists of three matrices—position, velocity, and acceleration—which may be missing. These matrices are defined for each individual road, and for roads of varying lengths the data are padded to ensure consistent input dimensions. The intuition is that vehicle states are not independent: traffic flow patterns, car-following behavior, and acceleration trends create strong spatial-temporal

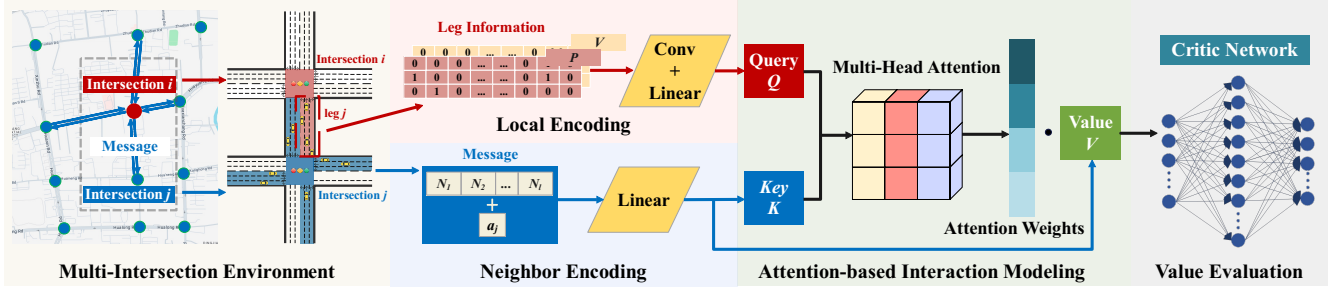


Fig. 8: Attention Module

correlations. For instance, if a vehicle suddenly decelerates while the preceding vehicle—at a visible distance—is still accelerating, it may indicate the presence of an unobserved vehicle in between. The reconstruction module leverages such correlations to infer missing data.

The network adopts a hierarchical encoder–decoder architecture that incorporates residual learning [37]. In the encoder stage, stacked convolutional layers and residual blocks extract localized spatial–temporal correlations among vehicles. These residual connections facilitate stable training and enable the network to capture behavioral patterns such as car-following dynamics, lane interactions, and acceleration trends. By progressively increasing feature dimensions, the encoder compresses the partially observed data into a compact latent representation that encodes both observed and inferred traffic conditions.

In the decoder stage, the latent representation is upsampled and then passed through fully connected layers to reconstruct the vehicle state matrices. Specifically, the network outputs two parallel components: a binary matrix $\hat{P}_i \in [0, 1]^{L \times N}$ representing the reconstructed vehicle positions, and a floating-point matrix $\hat{V}_i \in \mathbb{R}^{L \times N}$ representing the reconstructed velocities. Both outputs are refined through masking operations, where observed entries in the input are preserved, and empty positions are filled with values inferred by the model.

During training, the reconstructed position and velocity matrices are compared with the ground-truth ones from simulation, and the network is optimized by minimizing a hybrid reconstruction loss. Specifically, Binary Cross-Entropy (BCE) loss is applied for binary position recovery, while Mean Squared Error (MSE) is used for continuous velocity estimation. The overall loss $\mathcal{L}$ is a weighted combination of the two components:

$$\mathcal{L} = \lambda_1 \cdot \text{BCE}(y_{\text{pos}}, \hat{y}_{\text{pos}}) + \lambda_2 \cdot \text{MSE}(y_{\text{vel}}, \hat{y}_{\text{vel}}) \quad (15)$$

where λ_1 and λ_2 denote the balancing coefficients for position and velocity reconstruction, respectively.

During inference, the reconstructed states replace missing vehicle information and are fed into the actor–critic network, enabling the reinforcement learning agent to perform robust traffic signal control under low penetration rates of connected vehicles. Among the reconstructed features, the state attributes (Q_{ds} and H) are derived from the reconstructed position matrix, where the headway is estimated as the segment length minus the vehicle length.

By explicitly modeling spatial–temporal dependencies and completing missing data, the Scenario Reconstruction Module significantly improves the reliability of state representations in dynamic traffic environments.

E. Attention Module

To improve the efficiency and scalability of the critic network, an attention mechanism is employed to selectively focus on neighboring intersections. Unlike approaches that treat all agents equally, the mechanism prioritizes neighbors with greater influence on local signal control, incorporating their observations and actions into the value function. This design reduces the state space, lowers computational complexity, and improves learning efficiency as the number of agents increases.

Typically, the flow on each leg of an intersection originates from directly connected intersections. For example, in Fig. 8, the vehicles on the red leg of intersection i come from the incoming (blue) road of neighboring intersection j . Therefore, in order to evaluate the local policy within a dynamic surroundings, the critic network receives messages m from neighboring intersections and calculates the attention weights of the surrounding intersections by the Attention Module. The Q-value function $Q_i^\mu(s_i, a_i)$ for local agent i is represented in Eq. 16.

$$Q_i^\mu(s_i, a_i) = f_i(s_i, a_i, x_i) \quad (16)$$

f_i is a value function computed by critic network, x_i is the output of the Attention Module, which determines the importance of surrounding intersections. This mechanism allows the local agent to focus on the most relevant agents in its vicinity.

Within the Attention Module, feature processing follows the local and neighbor encoding stages shown in Fig. 8. For each agent i , the position and velocity features of each leg leg_j are processed by the local encoding block, while the message m_j —containing the vehicle counts N_l and action a_j of neighboring intersection j —is processed by the neighbor encoding block. These streams are passed through separate linear encoding layers to obtain latent representations.

The encoded leg features are mapped to the Query vector, and the encoded message features to the Key and Value vectors, yielding Q_j , K_j , and V_j as defined in Eq. 17. Here, j denotes a neighboring intersection and its corresponding connected leg of intersection i .

$$\begin{cases} Q_j = W_q \cdot leg_j \\ K_j = W_k \cdot m_j \\ V_j = W_v \cdot m_j \end{cases} \quad (17)$$

where W_q , W_k and W_v are the weight matrices for the Query, Key, and Value respectively. Then, Q_j , K_j , and V_j are split according to the number of attention heads.

For a neighboring intersection j , the attention weight is computed as Eq. 18.

$$\text{attn_weight}_j = \text{softmax}\left(\frac{Q_j \cdot K_j^T}{\sqrt{d_k}}\right) \quad (18)$$

where d_k is the dimension of the key vector.

Finally, the value outputs from each head are weighted and summed as Eq. 19, and the outputs attn_output_j from all heads are concatenated. The concatenated result is then passed through a linear layer to produce the final output x_i , serving as input to the critic network to compute the local Q-value as defined in Eq. 16.

$$\text{attn_output}_j = \text{attn_weight}_j \cdot V_j \quad (19)$$

V. EXPERIMENTS

In this section, we evaluate our method on four datasets and compare it with previous work.

A. Settings

1) *Experimental Environment*: The experiments are conducted on SUMO [38], a comprehensive traffic simulation platform that supports intermodal simulation, including pedestrian movements. Furthermore, all experiments are implemented in Python and executed on a system equipped with an Intel Core i9-10980XE CPU (3.00 GHz) and an NVIDIA GeForce RTX 3090 GPU.

2) *Parameter Configuration*: The hyperparameter configurations used in this study are summarized in Table III. These settings largely follow established practices from prior MADDPG-based studies [5], [39], [40]. Specifically, the learning rates for the actor and critic networks are typically in the range of 10^{-3} to 10^{-4} , the discount factor γ is chosen from $\{0.9, 0.95, 0.99\}$, and the sample size s is either 32 or 64, ensuring stable convergence and smooth policy updates. Remaining hyperparameters are empirically tuned for the traffic control scenarios in this study.

The reward function coefficients are determined based on the relative importance of vehicles and pedestrians. Waiting time serves as the baseline, with a weight of 1 or 2. For pedestrians, the impact of an additional second of waiting is smaller compared to vehicles, because vehicles typically form longer queues and generate greater congestion. Thus, the pedestrian weight β is selected from $[0.5, 1, 1.5]$. To balance encouraging more vehicles to pass and limiting excessive waiting, the coefficient σ for passed vehicles is chosen from $[1, 1.5, 2]$. Several configurations were tested, with detailed experiments in Appendix A. The final values in Table IV are selected based on convergence speed and overall performance during training.

TABLE III: Parameters Setting of DRATSC

Parameter	Symbol	Value
Training episodes	e	200
Sample size	s	64
Memory length	l	10000
Simulation time of each episode	t	600
Actor learning rate	$actor_{lr}$	1×10^{-3}
Critic learning rate	$critic_{lr}$	1×10^{-4}
Hidden layer dimension	$hidden_{dim}$	64
Attention extraction dimension	att_{dim}	64
Number of attention heads	$head_{num}$	4
Discount factor	γ	0.95
BCE loss weight	λ_1	2.0
MSE loss weight	λ_2	1.0

TABLE IV: Reward Coefficients

Weight for	Vehicles' Waiting Time	Pedestrians' Waiting Time	The Number of Passed Vehicles
Symbol	α	β	σ
Value	2	1	1.5

3) *Datasets*: In order to comprehensively evaluate the effectiveness of the proposed method, we conduct experiments on both synthetic and semi-synthetic datasets. The details of each dataset are as follows.

Synthetic_{2x3}: This dataset is self-constructed for this study and represents a road network of six intersections, each covering an area of 400 m $\times$ 600 m. During the 600-second simulation, 470 vehicles are randomly generated with randomly assigned routes, resulting in an uneven distribution. The vehicle arrival rate is set to one per second. To simulate weather-induced changes in driving behavior, the first 200 seconds are “Sunny” and the remaining 400 seconds are “Rainy”, with behavior variations detailed in Table V. Here, $v_{\max}$ denotes the maximum speed limit; v_{desire} , the drivers’ desired speed while ensuring safety; acc , the common acceleration; dec , the common deceleration; and τ , the time headway between vehicles. Parameter settings follow [41].

Jinan_{3x4}: This dataset¹ represents real-world traffic flow in Dongfeng Subdistrict, Jinan, China, with a road network of 12 intersections (each 400 m $\times$ 800 m). Vehicle routes are generated based on traffic records from surveillance cameras, including 6295 vehicles over one hour, with a pedestrian-to-vehicle ratio of 1:1. Due to privacy concerns regarding pedestrian routes, realistic movement patterns are derived from Unmanned Aerial Vehicle (UAV)-collected data at two intersections in Zhengzhou’s Gaoxin District. The resulting dataset represents typical urban traffic conditions.

Hangzhou_{4x4}: This dataset¹ originates from Gudang Subdistrict, Hangzhou, China, and models a 16-intersection network (each 600 m $\times$ 800 m). It includes 2,983 vehicle routes extracted from surveillance records and 3,952 pedestrian routes generated using the same UAV-based method as in the Jinan dataset, maintaining a realistic 1.325:1 pedestrian-to-vehicle ratio. Compared with the Jinan dataset, it features lower overall traffic volume, serving as a benchmark for evaluating model performance under relatively sparse conditions.

¹<https://github.com/nevesaynever1/datasetsforCAMABRL>

Hangzhou_{4×4}^{Syn}: Based on the real Hangzhou dataset¹, we develop a semi-synthetic variant to evaluate model performance under scalable traffic conditions. Vehicle and pedestrian departures follow Poisson processes with mean intervals of $\lambda_{veh} = 0.8$ and $\lambda_{ped} = 1.2$, capturing the stochastic and bursty characteristics of real traffic flows [42]. To represent heterogeneous road usage, each route is assigned a random weight within $[0.5, 1.5]$. This configuration produces 5,000 vehicles and 2,000 pedestrians per hour, maintaining the realistic road network while enabling systematic evaluation of model scalability and adaptability.

TABLE V: Different Driving Behaviors under Different Weather

Parameter	Sunny	Rainy
v_{max}	13.89 m/s	13.89 m/s
v_{desire}	13.89 m/s	9.72 m/s
acc	2.5 m/s ²	1.5 m/s ²
dec	5 m/s ²	4 m/s ²
τ	1.5 s	1 s

4) *Baselines*: To comprehensively evaluate the performance of our method, we compare it with five representative baselines that cover a wide spectrum of traffic signal control paradigms, including classical rule-based, adaptive, and DRL-based approaches. The key differences among baselines are summarized in Table VI.

Fixed Time [43]: A classical rule-based baseline that operates on fixed signal cycles. Its timing plans are predetermined based on historical traffic data, regardless of real-time traffic fluctuations.

SOTL [15]: A classical adaptive baseline that adjusts green time in response to real-time vehicle arrivals detected by sensors. While it is adaptive, it remains non-learning-based, representing a lightweight adaptive reference.

MADDPG [40]: A baseline that, similar to our approach, applies the MADDPG algorithm to control multiple intersections, modeling each intersection as an agent. The key difference is that MADDPG employs centralized training based on the global state of all intersections.

MARDDPG [5]: An extension of the MADDPG baseline that introduces LSTM into the framework. By introducing recurrent structures, it alleviates the instability caused by partial observability and dynamic traffic conditions, enabling agents to make more consistent decisions over time.

IPDALight [26]: An advanced DRL-based baseline that, like our approach, explicitly accounts for coordination among neighboring intersections. Built upon a DQN framework rather than MADDPG, it employs multi-scale intensity representation to enable adaptive phase timing and enhance GreenWave coordination.

5) *Metrics*: The evaluation metrics are similar to those adopted in [5], [44], and are computed directly from SUMO simulation data, including travel time of vehicles and waiting time of vehicles and pedestrians.

Average Waiting Time of Vehicles: The mean cumulative duration that vehicles remain stationary at intersections during their journeys.

$$\bar{W}_{veh} = \frac{\sum_{i=1}^{N_v} W_i}{N_v} \quad (20)$$

where W_i is the waiting time of vehicle i , defined as the cumulative duration during which its speed is below or equal to 0.1 m/s, and N_v is the number of vehicles that entered the road network.

Average Travel Time of Vehicles: The average time required for vehicles to complete their journeys.

$$\bar{T}_{veh} = \frac{\sum_{i=1}^{N_{vc}} T_i}{N_{vc}} \quad (21)$$

where T_i is the travel time of vehicle i , and N_v is the number of vehicles that completed their routes.

Average Waiting Time of Pedestrians: The mean duration pedestrians wait at intersections before crossing.

$$\bar{W}_{ped} = \frac{\sum_{j=1}^{N_p} W_j}{N_p} \quad (22)$$

where W_j is the waiting time of pedestrian j , and N_p is the number of pedestrians entered the road network.

As all evaluation criteria are temporal metrics, lower experimental results denote better performance.

B. Experiment Results

We conduct seven experiments across four scenarios under two different action spaces (two-phase and four-phase), with each simulation lasting 600 seconds. For the *Hangzhou_{4×4}^{Syn}* dataset under the two-phase setting, the simulation duration is extended to 1200 seconds to evaluate the model's performance on a larger network and over a longer simulation period. For the *Hangzhou_{4×4}* dataset, only four-phase experiments are conducted to examine the model's adaptability under low-traffic conditions. To ensure a fair evaluation of the intrinsic performance of the models, penetration rates are not considered in this section, and all data are assumed to be fully observable. Moreover, as MARDDPG fails to converge with 16 intersections in the four-phase scenario, it is modified to rely only on the historical information of adjacent intersections.

1) *Experiments in Two-phase Signal*: Table VII compares the performance of different methods under three traffic scenarios. It can be observed that DRATSC demonstrates strong performance across most metrics in all three scenarios.

In *Synthetic_{2×3}*, driving preferences change after 200 seconds to simulate varying weather, causing larger training fluctuations (Fig. 9a). IPDALight and DRATSC remain relatively stable, likely due to enhanced state representations: IPDALight captures fine-grained vehicle positions and speeds, while DRATSC additionally incorporates headway distances and downstream traffic states, better modeling interactions and adapting to behavior changes.

In *Jinan_{3×4}*, DRATSC achieves the best vehicle waiting time, though pedestrian waiting is higher than MADDPG. This may stem from the two-phase control, where shared phases for vehicles and pedestrians create a trade-off between vehicle efficiency and pedestrian convenience. In addition,

TABLE VI: Feature Comparison of Different Traffic Signal Control Methods

Method	Adaptive	RL-based	MADDPG-based	Global Coordination	Neighbors Coordination	Partial Observability	Pedestrian Awareness
Fixed Time [43]	×	×	×	×	×	×	×
SOTL [15]	✓	×	×	×	×	×	×
MADDPG [40]	✓	✓	✓	✓	×	×	×
MARDDPG [5]	✓	✓	✓	✓	×	✓	✓
IPDALight [26]	✓	✓	×	×	✓	×	×
DRATSC (ours)	✓	✓	✓	×	✓	✓	✓

* As a classical pre-timed and non-adaptive baseline, Fixed Time control inherently lacks all the listed advanced features.

TABLE VII: Comparison Results of Different Methods in Two-phase Signals

Method	Travel Time (s)			Waiting Time (s)			Pedestrians Waiting Time (s)	
	<i>Synthetic</i> _{2×3}	<i>Jinan</i> _{3×4}	<i>Hangzhou</i> _{4×4} ^{syn}	<i>Synthetic</i> _{2×3}	<i>Jinan</i> _{3×4}	<i>Hangzhou</i> _{4×4} ^{syn}	<i>Jinan</i> _{3×4}	<i>Hangzhou</i> _{4×4} ^{syn}
Fixed Time [43]	228.98 ± 0.24	224.99 ± 0.43	285.80 ± 0.77	47.69 ± 0.24	43.37 ± 0.13	48.27 ± 0.19	10.64 ± 0.01	5.64 ± 0.01
SOTL [15]	210.05 ± 0.27	217.18 ± 0.47	276.26 ± 0.18	35.48 ± 0.22	37.42 ± 0.11	35.82 ± 0.04	9.08 ± 0.01	8.50 ± 0.01
MADDPG [40]	194.79 ± 0.34	214.77 ± 0.26	270.89 ± 0.33	26.78 ± 0.26	28.59 ± 0.13	26.28 ± 0.16	2.20 ± 0.01	3.53 ± 0.02
MARDDPG [5]	194.18 ± 0.35	215.49 ± 0.71	267.02 ± 0.27	25.86 ± 0.26	31.97 ± 0.18	21.51 ± 0.16	3.35 ± 0.02	3.73 ± 0.03
IPDALight [26]	181.59 ± 0.33	211.18 ± 0.47	263.26 ± 1.50	19.81 ± 0.16	25.55 ± 0.19	18.72 ± 0.07	2.26 ± 0.01	3.28 ± 0.04
DRATSC(ours)	180.46 ± 0.32	212.11 ± 0.60	260.65 ± 0.16	17.66 ± 0.17	24.39 ± 0.19	16.94 ± 0.15	2.48 ± 0.02	3.13 ± 0.05

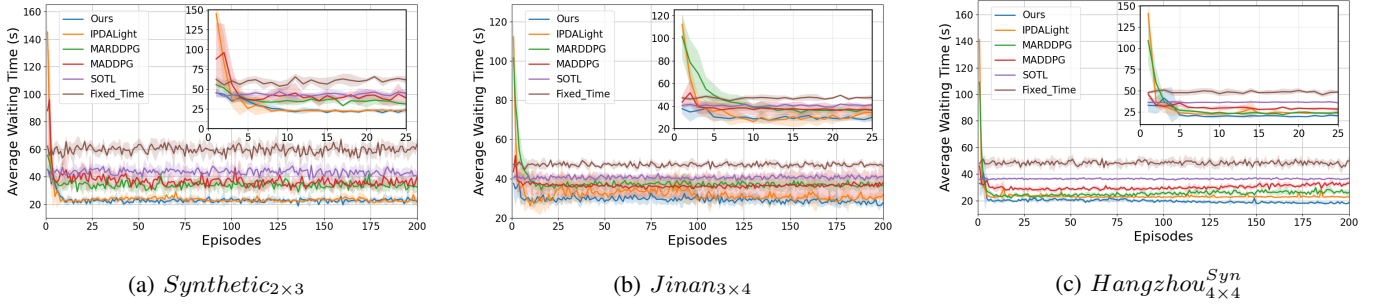


Fig. 9: Average Waiting Time of Vehicles in Different Scenarios (Two-phase)

its travel time is higher than that of IPDALight, possibly because the travel time metric only accounts for vehicles that have completed their trips, while more unfinished trips remain waiting on the map.

In the *Hangzhou*_{4×4}^{syn} scenario, which consists of 16 intersections, DRL-based methods exhibit a more pronounced performance advantage over traditional non-DRL-based traffic signal control approaches. This highlights the significant potential of intelligent traffic light systems that leverage in large-scale road network.

Additionally, by examining the three graphs in Fig. 9, it is evident that all DRL-based methods converge rapidly and exhibit stable performance. This can be largely attributed to the reduced action space associated with two-phase control.

In summary, under two-phase control, our method demonstrates an overall improvement of 50% compared to traditional approaches and achieves approximately 8% better performance than IPDALight in terms of vehicles waiting time.

2) *Experiments in Four-phase Signal*: We evaluate four-phase traffic signal control across four distinct scenarios, with our method achieving superior overall performance. The analysis is organized along four key aspects: the impact of phase design, the influence of underlying model architectures, the performance under low traffic flow, and pedestrian-awareness in control design.

As shown in Table VIII, increasing the number of signal phases from two to four leads to a two- to three-fold rise in average vehicle waiting time across most scenarios. Pedestrian waiting time also increase, though the growth is less significant in methods explicitly considering pedestrians, such as MARDDPG and DRATSC. The overall degradation mainly results from separating left-turn and through movements, which reduces intersection throughput and limits pedestrian crossing opportunities during left-turn phases.

As shown in Fig. 10, convergence under four-phase control is generally slower than under two-phase control across most scenarios. This is especially evident in the *Hangzhou*_{4×4}^{syn} scenario, where models such as IPDALight, MARDDPG, and ours only begin to converge after approximately 100 episodes, likely due to its more complex action space and higher number of intersections.

Beyond convergence speed, the training dynamics highlight the stability advantage of the proposed method. In the *Synthetic*_{2×3}, *Jinan*_{3×4}, and *Hangzhou*_{4×4}^{syn} scenarios, IPDALight shows more pronounced oscillations due to its independent DQN-based learning, which can cause non-stationarity in multi-agent settings. In contrast, our approach maintains stable convergence by leveraging inter-agent coordination through a centralized critic, effectively capturing joint dependencies among intersections and facilitating cooperative

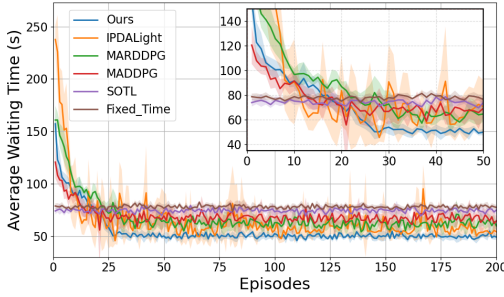
TABLE VIII: Comparison Results of Different Methods in Four-phase Signals

(a) Vehicle Performance Comparison

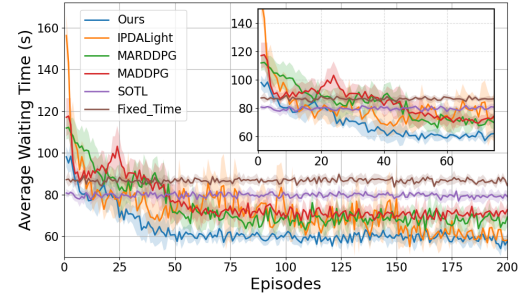
Method	Travel Time (s)				Waiting Time (s)			
	$Synthetic_{2 \times 3}$	$Jinan_{3 \times 4}$	$Hangzhou_{4 \times 4}$	$Hangzhou_{4 \times 4}^{syn}$	$Synthetic_{2 \times 3}$	$Jinan_{3 \times 4}$	$Hangzhou_{4 \times 4}$	$Hangzhou_{4 \times 4}^{syn}$
Fixed Time [43]	223.30 $\pm$ 0.19	250.86 $\pm$ 0.14	256.51 $\pm$ 0.27	235.68 $\pm$ 0.31	75.52 $\pm$ 0.06	82.44 $\pm$ 0.29	41.38 $\pm$ 0.45	79.23 $\pm$ 0.19
SOTL [15]	220.65 $\pm$ 0.13	247.33 $\pm$ 0.49	253.40 $\pm$ 0.25	230.84 $\pm$ 0.26	71.41 $\pm$ 0.17	75.13 $\pm$ 0.12	35.16 $\pm$ 0.45	72.57 $\pm$ 0.09
MADDPG [40]	210.47 $\pm$ 0.24	239.89 $\pm$ 0.87	244.21 $\pm$ 0.11	229.38 $\pm$ 1.02	61.76 $\pm$ 2.57	63.80 $\pm$ 0.73	26.94 $\pm$ 0.79	61.55 $\pm$ 0.50
MARDDPG [5]	206.70 $\pm$ 0.33	233.16 $\pm$ 0.34	232.78 $\pm$ 0.10	230.70 $\pm$ 0.28	57.23 $\pm$ 0.85	58.90 $\pm$ 0.32	16.98 $\pm$ 0.15	59.37 $\pm$ 0.18
IPDALight [26]	192.67 $\pm$ 0.26	229.63 $\pm$ 1.01	233.42 $\pm$ 0.12	222.05 $\pm$ 0.77	49.98 $\pm$ 0.85	53.98 $\pm$ 0.54	15.67 $\pm$ 0.35	49.16 $\pm$ 0.85
DRATSC (ours)	189.19 $\pm$ 0.20	228.81 $\pm$ 0.26	232.76 $\pm$ 0.19	220.08 $\pm$ 1.83	47.45 $\pm$ 0.10	51.23 $\pm$ 0.38	15.80 $\pm$ 0.26	48.48 $\pm$ 0.97

(b) Pedestrians Performance Comparison

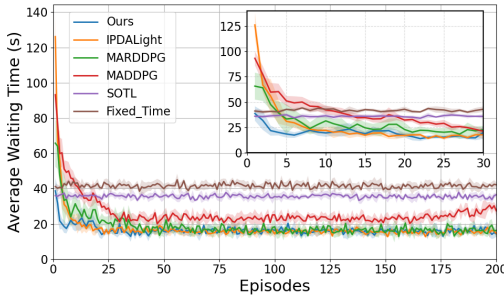
Method	Pedestrian Waiting Time (s)		
	$Jinan_{3 \times 4}$	$Hangzhou_{4 \times 4}$	$Hangzhou_{4 \times 4}^{syn}$
Fixed Time [43]	21.24 $\pm$ 0.04	8.45 $\pm$ 0.04	22.63 $\pm$ 0.03
SOTL [15]	21.97 $\pm$ 0.02	8.89 $\pm$ 0.02	17.78 $\pm$ 0.06
MADDPG [40]	11.50 $\pm$ 0.10	3.51 $\pm$ 0.02	11.21 $\pm$ 0.12
MARDDPG [5]	4.24 $\pm$ 0.01	2.96 $\pm$ 0.03	3.30 $\pm$ 0.01
IPDALight [26]	8.79 $\pm$ 0.20	7.84 $\pm$ 0.04	8.28 $\pm$ 0.14
DRATSC (ours)	3.33 $\pm$ 0.02	2.48 $\pm$ 0.09	3.21 $\pm$ 0.11



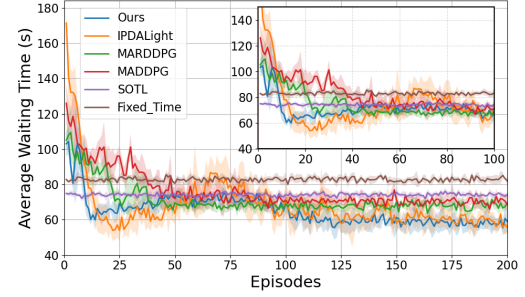
(a) $Synthetic_{2 \times 3}$



(b) $Jinan_{4 \times 4}$



(c) $Hangzhou_{4 \times 4}$



(d) $Hangzhou_{4 \times 4}^{Syn}$

Fig. 10: Average Waiting Time of Vehicles in Different Scenarios (Four-phase)

policy learning.

It is worth noting that the $Hangzhou_{4 \times 4}$ scenario exhibits a distinct pattern, as illustrated in Fig. 10c. In this scenario, all models converge rapidly and remain stable after early training. This behavior is likely related to the lower overall traffic volume and the highly regular east–west–dominated flow distribution. Such a relatively simple and less congested traffic pattern leads to a reduced decision space and diminishes the need for complex inter-intersection coordination. In addition, a noticeable performance gap persists between traditional rule-based methods and DRL-based approaches,

indicating that learning-based models are more effective at capturing underlying traffic regularities, especially in scenarios with pronounced spatial or directional flow patterns.

The importance of pedestrian-aware design is further corroborated by real-time pedestrian waiting statistics in the $Hangzhou_{4 \times 4}^{Syn}$ scenario (Fig. 11). Models that explicitly account for pedestrians, such as MARDDPG and DRATSC, consistently maintain low waiting numbers, with peak values not exceeding 20 pedestrians per second. In contrast, pedestrian-unaware methods like MADDPG and IPDALight exhibit periodic surges and a sharp increase toward the end of

the simulation.

In summary, under four-phase control, our method achieves an overall 40% reduction in average waiting time compared to traditional methods and outperforms IPDALight in most scenarios, demonstrating its effectiveness in balancing vehicle and pedestrian demands in complex traffic environments.

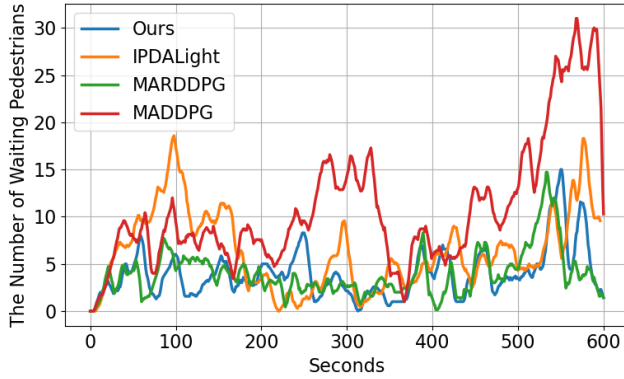


Fig. 11: Real-Time Number of Waiting Pedestrians Across Models

3) *Training Stability and Convergence:* In this experiment, we assess the training stability and convergence of the proposed model under similar traffic demand but different stochastic flow conditions. To achieve this, SUMO's random seed mechanism is adopted, where the overall traffic intensity is determined by fixed configuration parameters, while the random seed affects the generation of vehicles and pedestrians, including their departure times, routes, and spatial distributions.

As shown in Fig. 12, all training curves within the same scenario converge to similar reward levels, indicating that the proposed model achieves stable and consistent learning behavior despite randomness in traffic generation. In contrast, the clear differences between the $Jinan_{3 \times 4}$ and $Hangzhou_{4 \times 4}$ scenarios reflect the influence of distinct network structures and traffic complexities on convergence dynamics.

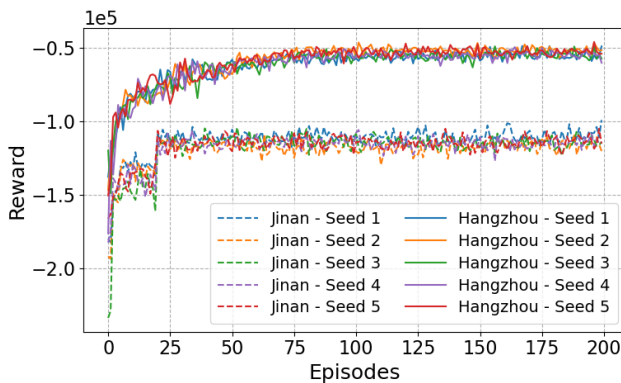


Fig. 12: Reward of DRATSC in Jinan and Hangzhou Scenarios

C. Ablation Study

In order to validate the effectiveness of different modules in our proposed approach, we conduct ablation experiments focusing on three key aspects: the Scenario Reconstruction Module, the Attention Module and pedestrian awareness.

1) Experiments on Scenario Reconstruction Module:

To evaluate the reconstruction performance under different VANETs penetration rates, three metrics—Precision, Recall, and F1-score—are employed. Their calculation formulas and detailed results are presented in Appendix B. Here, F1-score serves as the primary metric for evaluating the accuracy and completeness of traffic state recovery, with higher values indicating better reconstruction performance.

Following Zhang et al. [3], the V2I penetration rate p is determined by a Bernoulli process, wherein each vehicle is independently assigned as detected or undetected with probability p . Detected vehicles provide real-time state data to the model, whereas the states of undetected vehicles remain unobservable and are recovered by the reconstruction module. The penetration rate p varies from 10% to 90% in increments of 10%.

As shown in Fig. 13, the F1-score increases steadily with the rise of V2I penetration rate in both two scenarios. At a low penetration rate of 10%, the F1-scores remain at 0.55 ($Jinan_{3 \times 4}$) and 0.47 ($Hangzhou_{4 \times 4}^{Syn}$), indicating that the module can already partially recover missing traffic states even under highly sparse observations. As the penetration rate rises to 40–60%, the F1-scores improve significantly to above 0.8, demonstrating the model's ability to effectively infer unobserved vehicles and maintain high reconstruction accuracy. However, beyond this range, the improvement slows down, indicating that adding more data brings smaller gains as the system nears its performance limit.

Moreover, both the $Jinan_{3 \times 4}$ and $Hangzhou_{4 \times 4}^{Syn}$ scenarios exhibit satisfactory and comparable reconstruction performance, indicating consistent reconstruction capability and strong generalization across different urban scenarios.

To further assess the model performance under different V2I penetration levels, comparative experiments are conducted on multiple benchmark methods, including our proposed model with and without the reconstruction module. Each penetration setting is tested ten times, and the average vehicle waiting time is reported.

As shown in Fig. 14, the performance gap between the models with (blue) and without (orange) the reconstruction module is evident, particularly under low penetration conditions. This indicates that the module effectively compensates for missing observations and enhances robustness in partially observable environments.

Meanwhile, different baseline models exhibit distinct sensitivities to V2I penetration rates. MARDDPG maintains relatively strong performance at moderate penetration levels (30%–60% in Jinan), comparable to our model without the reconstruction module, indicating that its LSTM mechanism partially mitigates the impact of sparse observations. In contrast, IPDALight shows a sharp performance drop under low penetration (10%–30%) due to its strong reliance on directly observed vehicle data.

Overall, these results demonstrate that the proposed reconstruction module enhances adaptability and stability, enabling reliable decision-making under varying V2I penetration.

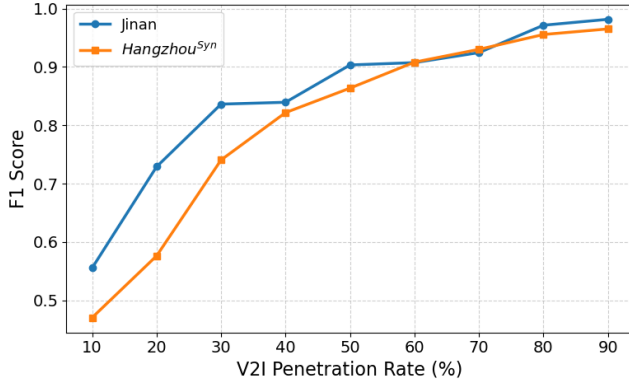
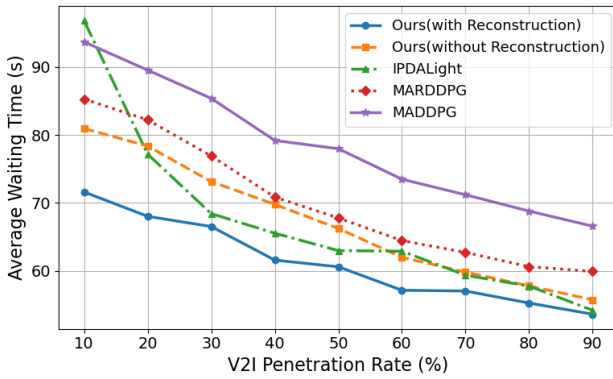
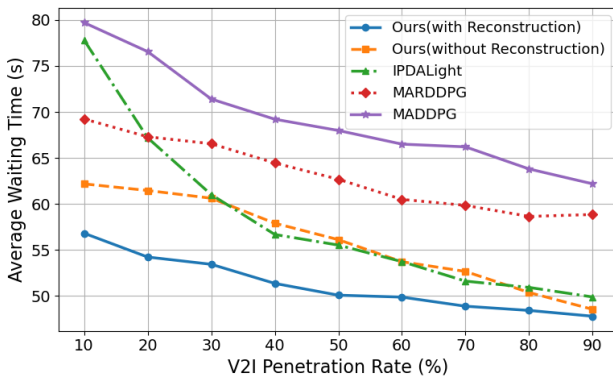


Fig. 13: Impact of V2I Penetration Rate on Model Performance



(a) $Jinan_{3 \times 4}$



(b) $Hangzhou_{4 \times 4}^{Syn}$

Fig. 14: Average Waiting Time of Vehicles With and Without Reconstruction

2) *Experiments on Attention Module:* To investigate the contribution of the attention mechanism, we conduct an ablation study in the $Hangzhou_{4 \times 4}^{Syn}$ scenario. In this study, one variant of the critic network directly takes messages from neighboring intersections as input without any weighting,

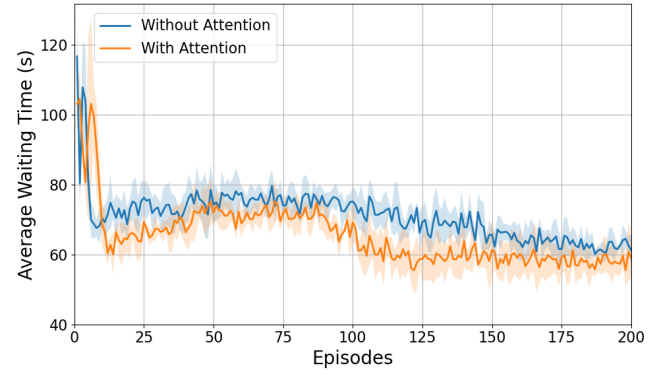


Fig. 15: Comparison With and Without Attention Module

while the other variant processes these messages through the Attention Module to compute weighted contributions from each neighbor. As shown in Fig. 15, incorporating the Attention Module leads to a clear improvement in overall performance, verifying its effectiveness in capturing the most relevant inter-intersection interactions.

3) *Experiments on Pedestrian Observation:* To validate the effectiveness of pedestrian-aware optimization, we conduct comparative experiments under two settings: pedestrian-unaware and pedestrian-aware scenarios. In the pedestrian-aware setting, pedestrian states are incorporated into the model inputs, while they are omitted in the pedestrian-unaware setting. The average pedestrian waiting time is reported in Table IX. The results show that incorporating pedestrian awareness enables the model to make decisions that better account for and reduce pedestrian waiting time.

TABLE IX: Average Waiting Time with Pedestrians Aware vs. Unaware (s)

Scenario	$Jinan_{3 \times 4}$	$Hangzhou_{4 \times 4}$	$Hangzhou_{4 \times 4}^{Syn}$
Pedestrian-Unaware	5.60 ± 0.09	5.53 ± 0.14	7.11 ± 0.23
Pedestrian-Aware	3.33 ± 0.02	2.48 ± 0.09	3.21 ± 0.11

VI. DISCUSSION

Scalability. To evaluate the scalability of our method in road networks of different sizes, we conduct experiments on networks with 4, 6, 9, 12, and 16 intersections, while keeping the traffic flow consistent across all environments. The average waiting time of vehicles are shown in Fig. 16. Compared to the stable and scalable fixed-time method, both approaches experience an increase in waiting time when the number of intersections grows from 4 to 6. This can be attributed to the fact that in smaller networks (e.g., 4 intersections), vehicles generally traverse fewer intersections per trip, leading to shorter travel and waiting times. As the number of intersections increases, the performance of the fixed-time method remains largely unchanged, while our method stabilizes after 9 intersections, demonstrating its scalability to larger road networks.

This scalability stems from its decentralized design: each agent focuses solely on neighboring intersections rather than the entire network, which not only enhances learning efficiency but also enables adaptation to road networks of different sizes.

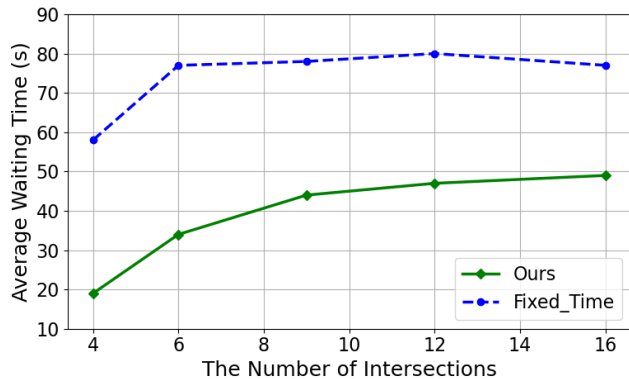


Fig. 16: Vehicles Waiting Time of DRATSC at Different Numbers of Intersections

Generalization. Although our model demonstrates strong scalability with respect to network size, it currently lacks the ability to handle heterogeneous intersection geometries (e.g., 3-way, 5-way, or more complex layouts). This limitation mainly arises from the reliance of both state inputs and action outputs on fixed dimensions, which restricts generalization across diverse intersection architectures.

Nevertheless, our model is naturally extensible through more flexible representations. On the state side, inspired by TransferLight [45], lane-level observations can be projected into a shared feature space and subsequently aggregated (e.g., via graph attention) into a unified encoding, thereby ensuring consistent representation across intersections with varying numbers of lanes. On the action side, an attention-based scoring mechanism can dynamically evaluate candidate phases, allowing the actor to generate probability distributions over variable-sized phase sets without requiring a fixed output dimension, akin to the principle adopted in AttendLight [46].

Furthermore, the Scenario Reconstruction Module already operates at the approach level, where padding can normalize differences in lane length or count. Combined with the attention-based critic network, which inherently aggregates information from a variable number of neighboring intersections by computing attention scores for each neighbor, these designs enable our framework to robustly generalize to heterogeneous geometries without altering the core learning paradigm.

Real-time performance. Since traffic signal control needs to respond quickly to dynamic traffic flow, the algorithm's inference speed is crucial. In the experiment, our method is able to complete an inference within 2ms, ensuring the feasibility of real-time adjustment. Besides, V2X communication, a key application of 5G technology, achieves millisecond-level latency. This ensures low latency and high reliability, making real-time vehicle data acquisition and optimized traffic control via VANETs both feasible and effective.

Robustness. Three key designs are implemented to enhance the robustness of our method: incorporating pedestrian impact in signal control, improving VANETs penetration through Scenario Reconstruction Module, and considering vehicle speed and headway distance to adapt to varying weather conditions. The experimental results demonstrate that these strategies effectively optimize model performance, ensuring robustness across different scenarios.

VII. CONCLUSION

In this paper, we present DRATSC, a novel method for intelligent multi-intersection traffic signal control. Our method processes the vehicle data obtained from VANETs to generate optimal traffic signal control strategies, effectively reducing average waiting time of vehicles and improving traffic flow. Considering the mixed traffic environment, our method explicitly addresses low V2I penetration through scenario reconstruction, ensures reasonable pedestrian waiting times, and adapts to weather-induced congestion patterns. Experimental results in the SUMO platform demonstrate that our method yields effective results across various scenarios and traffic conditions.

ACKNOWLEDGMENT

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APPENDIX A EXPERIMENTS ON REWARD FUNCTION PARAMETERS

To determine the optimal reward parameters, we conducted training experiments in the $Jinan_{3 \times 4}$ scenario using different combinations of (α, β, σ) . Each configuration was trained for 200 episodes with a duration of 300 seconds per episode, and the convergence curves of average waiting time are shown in Fig. 17.

Subsequently, three quantitative indicators were computed for each parameter set: 1) Final average performance, the mean over the last 25% of training episodes, indicating stable performance; 2) Stability, the relative fluctuation within the final 25% episodes, reflecting consistency; and 3) Convergence speed, the first epoch window maintaining performance within 20% of the final average for 30 consecutive episodes, measuring convergence rate.

The resulting heatmap in Fig. 18 visualizes the unified scores under different (α, β, σ) combinations, where higher scores indicate better convergence stability and efficiency. The results of selecting the top three parameter combinations, each evaluated through five independent inference runs, are summarized in Table X. Among all configurations, the parameter set $(\alpha, \beta, \sigma) = (2.0, 1.0, 1.5)$ achieves the best overall trade-off between vehicle efficiency and pedestrian waiting time, and is therefore selected as the final configuration.

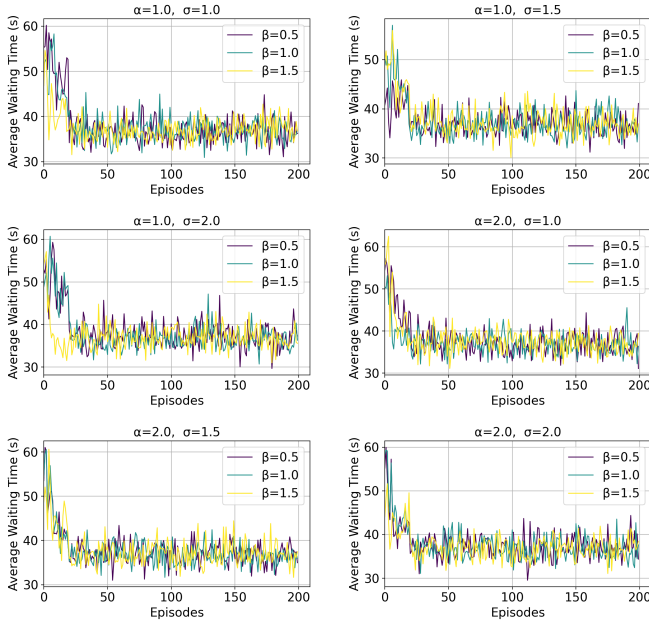


Fig. 17: Convergence Curves of Average Waiting Time under Different Reward Parameter Combinations

APPENDIX B ANALYSIS OF RECONSTRUCTION RESULTS

The reconstruction performance is evaluated using three standard classification metrics: *Precision*, *Recall*, and *F1*-

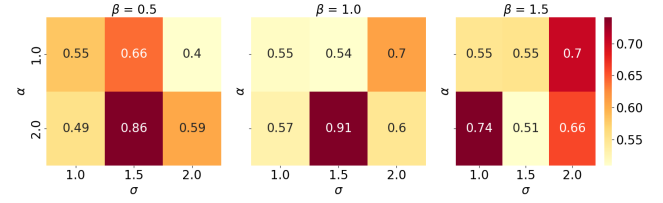


Fig. 18: Comprehensive Performance Visualization under Different Reward Parameter Combinations

TABLE X: Performance Comparison under Different Reward Parameter Combinations

α	β	σ	Travel Time (s)	Waiting Time (s)	Pedestrian Waiting Time (s)
2.0	0.5	1.5	150.79 ± 2.55	38.75 ± 0.29	2.91 ± 0.07
2.0	1.0	1.5	146.58 ± 1.36	36.89 ± 0.34	2.84 ± 0.13
2.0	1.5	1.0	151.46 ± 1.98	39.66 ± 0.63	2.66 ± 0.06

score, defined as follows:

$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} \\
 \text{Recall} &= \frac{TP}{TP + FN} \\
 \text{F1-score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}
 \end{aligned} \tag{23}$$

where TP , FP , and FN denote true positives, false positives, and false negatives, respectively. They are computed based on whether the presence of vehicles at each spatial position is correctly recovered in the reconstructed results.

Precision measures the proportion of correctly reconstructed vehicles among all predicted ones, while Recall evaluates how many real vehicles were successfully reconstructed.

As shown in Table XI, the Precision remains consistently high (above 0.97) across all penetration levels, indicating that the reconstruction module rarely produces false positives. Meanwhile, the Recall increases steadily with the V2I penetration rate, reflecting that richer observed information provides stronger contextual cues for inferring unobserved vehicles.

TABLE XI: Precision, Recall, and F1-Score under Different V2I Penetration Rates.

Penetration (%)	$Jinan_{3 \times 4}$			$Hangzhou_{4 \times 4}^{Syn}$		
	Precision	Recall	F1	Precision	Recall	F1
10	0.9820	0.3872	0.5554	0.9865	0.3087	0.4703
20	0.9787	0.5808	0.7290	0.9847	0.4072	0.5762
30	0.9838	0.7272	0.8363	0.9835	0.5939	0.7406
40	0.9828	0.7326	0.8394	0.9851	0.7048	0.8217
50	0.9865	0.8332	0.9034	0.9779	0.7733	0.8637
60	0.9855	0.8408	0.9074	0.9796	0.8465	0.9082
70	0.9876	0.8691	0.9246	0.9734	0.8906	0.9301
80	0.9890	0.9547	0.9715	0.9746	0.9373	0.9556
90	0.9902	0.9733	0.9817	0.9722	0.9583	0.9652